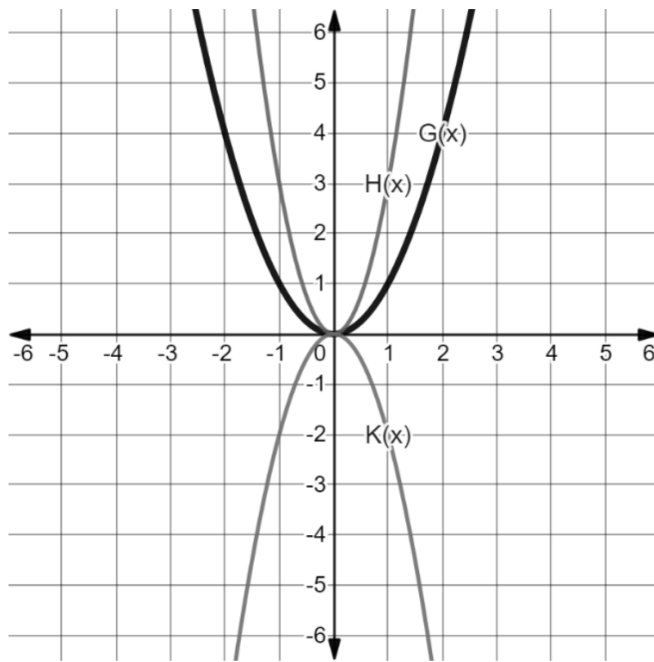


The functions  $G(x)$ ,  $H(x)$ , and  $K(x)$  are shown on the graph, where  $G(x) = x^2$ ,  $H(x) = 3x^2$ , and  $K(x) = -2x^2$ . The table of values for each function are also shown.



| $y = G(x)$ |     | $y = H(x)$ |     | $y = K(x)$ |     |
|------------|-----|------------|-----|------------|-----|
| $x$        | $y$ | $x$        | $y$ | $x$        | $y$ |
| -3         | 9   | -3         | 27  | -3         | -18 |
| -2         | 4   | -2         | 12  | -2         | -8  |
| -1         | 1   | -1         | 3   | -1         | -2  |
| 0          | 0   | 0          | 0   | 0          | 0   |
| 1          | 1   | 1          | 3   | 1          | -2  |
| 2          | 4   | 2          | 12  | 2          | -8  |
| 3          | 9   | 3          | 27  | 3          | -18 |

Complete the statements below.

- The coefficient of  $G(x) = x^2$  is \_\_\_\_\_.
- The coefficient of  $H(x) = 3x^2$  is \_\_\_\_\_.
- The  $y$ -values of  $H(x)$  can be found by multiplying the  $y$ -values of  $G(x)$  by \_\_\_\_\_.
- The coefficient of  $K(x) = -2x^2$  is \_\_\_\_\_.
- The  $y$ -values of  $K(x)$  can be found by multiplying the  $y$ -values of  $G(x)$  by \_\_\_\_\_.

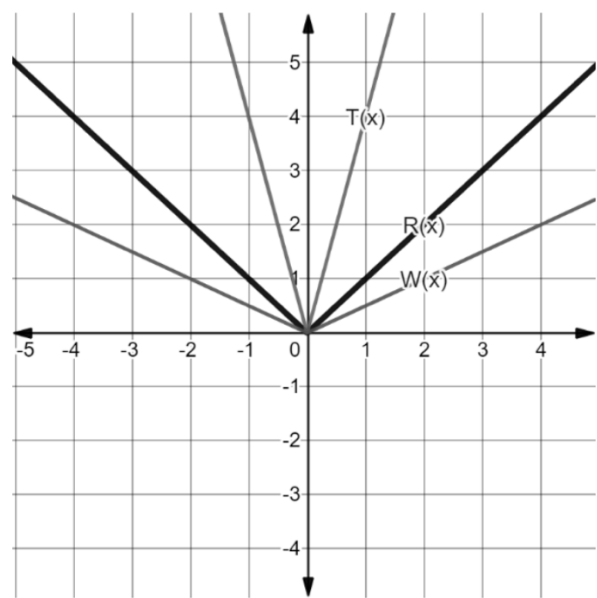
The table of values of the function  $y = A(x)$  is shown.

| $x$ | -6 | -4 | -2 | 0  | 2  | 4  | 6  |
|-----|----|----|----|----|----|----|----|
| $y$ | 4  | 7  | 10 | 13 | 16 | 19 | 22 |

Complete the sentence to describe the effect on the  $x$ -values for the function  $y = A(3x)$ .

6. The  $x$ -values for  $A(x)$  will be 
☐ multiplied  
☐ divided
  by 3.

The functions  $R(x)$ ,  $T(x)$ , and  $W(x)$  are shown on the graph, where  $R(x) = |x|$ ,  $T(x) = |4x|$ , and  $W(x) = |0.5x|$ . The table of values for each function are also shown.



| $y = R(x)$ |     | $y = T(x)$     |     | $y = W(x)$ |     |
|------------|-----|----------------|-----|------------|-----|
| $x$        | $y$ | $x$            | $y$ | $x$        | $y$ |
| -2         | 2   | $-\frac{1}{2}$ | 2   | -4         | 2   |
| -1         | 1   | $-\frac{1}{4}$ | 1   | -2         | 1   |
| 0          | 0   | 0              | 0   | 0          | 0   |
| 1          | 1   | $\frac{1}{4}$  | 1   | 2          | 1   |
| 2          | 2   | $\frac{1}{2}$  | 2   | 4          | 2   |

Complete the statements below.

7.

The  $x$ -values of  $T(x)$  can be found by \_\_\_\_\_.

☐ multiplying

☐ dividing

the  $x$ -values of  $R(x)$  by \_\_\_\_\_.
8.

The  $x$ -values of  $W(x)$  can be found by \_\_\_\_\_.

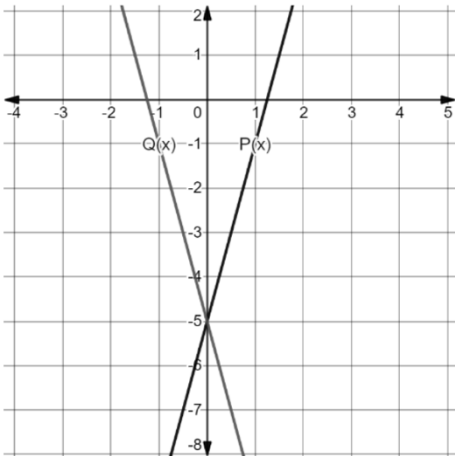
☐ multiplying

☐ dividing

the  $x$ -values of  $R(x)$  by \_\_\_\_\_.

The functions  $P(x)$  and  $Q(x)$  are shown on the graph where  $Q(x) = P(-x)$ . The table of values for each function are also shown.

| $y = P(x)$ |     | $y = Q(x)$ |     |
|------------|-----|------------|-----|
| $x$        | $y$ | $x$        | $y$ |
| -2         | -13 | -2         | 3   |
| -1         | -9  | -1         | -1  |
| 0          | -5  | 0          | -5  |
| 1          | -1  | 1          | -9  |
| 2          | 3   | 2          | -13 |



9.
- What is the effect on the table of values when the  $x$  in  $P(x)$  was replaced with  $-x$  to create  $Q(x)$ ?

10.

What is the effect on the graph when the  $x$  in  $P(x)$  was replaced with  $-x$  to create  $Q(x)$ ?